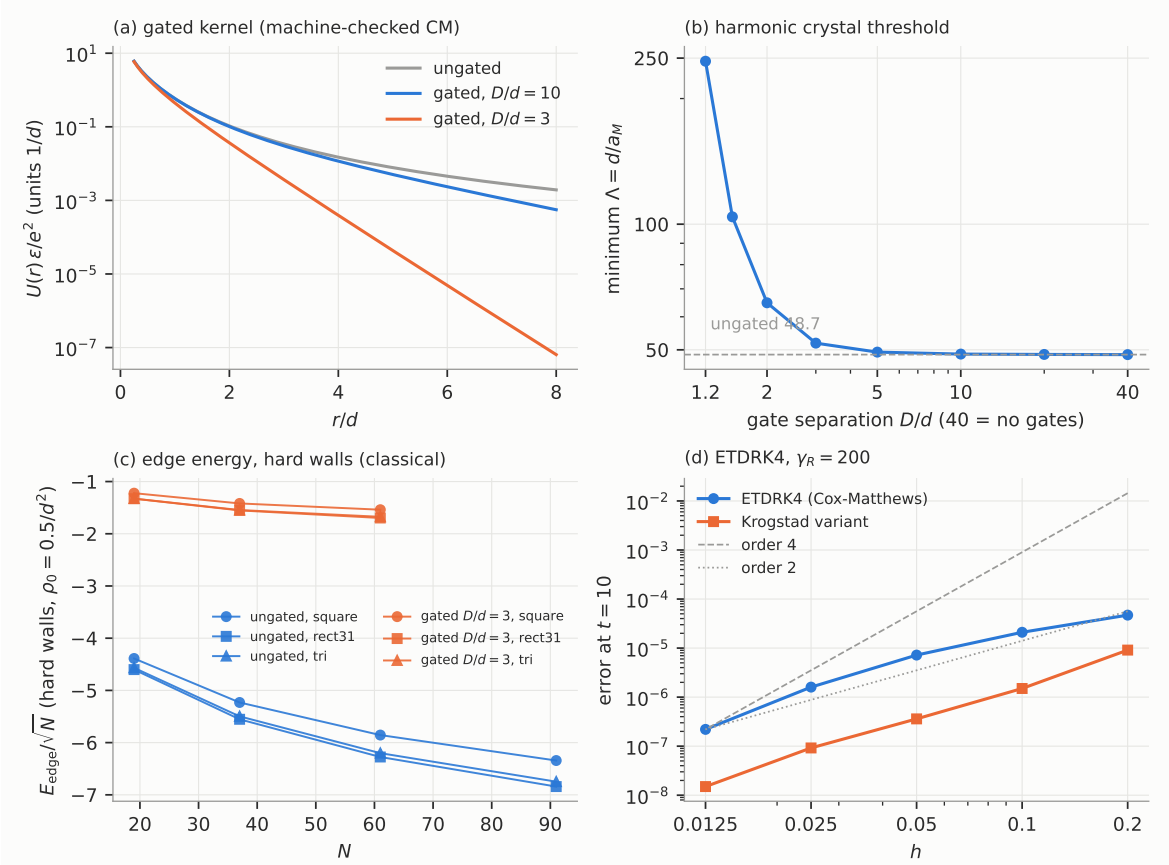


(a) Gated dipole kernel for two gate separations against the ungated kernel (Poisson series, $d = 1$). (b) Smallest $\Lambda = d/a_M$ admitting a harmonic-stable crystal against D/d (eight sampled values; 40 stands for no gates). (c) Edge energy per $\sqrt{N}$ of hard-wall containers, best configurations. (d) Error of ETDRK4 and of Krogstad's variant at the registered stiffness against step size; guide lines are anchored at the smallest step.

Gated exciton fluids II:

a machine-checked kernel theorem, a harmonic crystallisation estimate, open-boundary energies, a validity gate for materials-data inputs, and an exponential integrator that did not meet its gate

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Abstract

The first gated-exciton paper [1] proved on paper that grounded metal gates keep the dipolar exciton kernel completely monotone (CM) in r^2, ran a pre-registered driven-dissipative solver test, and listed what was not done. This second paper reports five of the follow-up packages. The four numerical packages (harmonic estimate, edge study, HF gate, ETD) carry a pre-registration committed before their registered runs (with the exceptions disclosed in Section 1); the Lean package has none. Each was read by an independent verifier before this deposit. **(1)** The theorem is formalised in Lean 4/Mathlib: 52 theorems, no `sorry`, only the standard three axioms, with the Poisson summation *derived* (not assumed) and the Laplace density shown non-negative and bounded; the verifier compiled it independently and recomputed the kernel by a direct image sum. **(2)** A harmonic Lindemann estimate of the crystallisation threshold, calibrated on a quantum-Monte-Carlo control, is a *negative*

result for devices with $d \sim 1$ nm: $d/a_M \gtrsim 49$ would be needed in a rigid-exciton model, gates raise it at every sampled gate separation, and two registered predictions on the control failed or were refuted (one of them depends on a definition in the QMC source that was not checked). **(3)** An open-boundary study of 65 classical N-particle configurations of three CM kernels finds a negative edge energy that depends monotonically on the aspect ratio (with one disclosed exception), and no clean boundary law. **(4)** A Hugging Face / neural-network validity gate *fails* for the two quantities the model needs. **(5)** An ETDRK4 integrator fails its registered order-4 gate on the stiff reservoir (consistent with stiff-order reduction), took 7.7 s against 227 s for dense BDF to a stationary state on one seed, 8×8 grid, on a loaded machine, and has a step limit lower than the IMEX module's at high density. Two packages, a matrix-free Newton–Krylov solver at 64×64 and a classical L6 node-arrangement design study, were still running at deposit time and are *not* reported and no verdict on them is implied. Written by an AI system; reviewed by independent instances; not refereed.

1 Status, scope, and what was and was not verified

Dependency. The lattice theorem of Cohn–Kumar type used throughout [2, 3] is unrefereed and AI-generated; every statement that uses it is conditional. Nothing here compares with experimental data. **Process.** Each numerical package was run by a worker instance with a pre-registration committed before the registered runs (dates and hashes in the repository), then re-checked by a fresh *hostile verifier* that recomputed key numbers with its own code. Exceptions: the Lean package has no pre-registration; in the HF package the one candidate data file was opened before the registration (disclosed there); the Wigner registration has amendment A2, written after the scan and marked exploratory; the edge study's later amendments are discussed in Section 4. Verifier verdicts: Lean *accept*; Wigner, edge, HF gate, ETD *accept with fixes*. The fixes are applied or disclosed below; claims the verifier could not reproduce were dropped.

Where a number appears, it is copied from a committed result file (named in Section 9); the figures are generated by `exploration/exciton/python/companion_figures.py` from those files. The only values typed into a figure script are the ETD work–precision rows, copied from the ETD results note and marked as such in the script.

2 Lean formalisation of the gated-kernel theorem

Statement. For grounded planes at $z = h_1$ and $z = -h_2$ ($P = 2(h_1 + h_2)$), an electron at z_e and a hole at z_h (any real values), the image series of $V_{ee} + V_{hh} - 2V_{eh}$ equals the Laplace transform $\int_0^\infty e^{-ts}\mu(s)\,ds$ of $\mu(s) = (\pi s)^{-1/2}S(s)$ with $S(s) \geq 0$; hence the kernel is CM in $t = r^2$.

What is proved (file `exploration/exciton/lean/GatedKernel.lean`, 1083 lines, 52 theorems, built against Mathlib v4.34.1; all 17 `#print axioms` show `[propext, Classical.choice, Quot.sound]`):

tier	content
T1	the trigonometric core $2 - \cos 2a - \cos 2b - 2\cos(b-a) + 2\cos(a+b) = 2(1-\cos(b-a))(1+\cos(a+b)) = 2(\sin a - \sin b)^2 \geq 0$; the image offsets on each plane are a bijective re-indexing of the direct ones, so the lattice Green function vanishes on both planes
T2	Poisson summation *derived* from Mathlib's Gaussian/Jacobi-theta results for every real shift; $S_m \geq 0$; $S(s) \geq 0$ for all $s > 0$, all $h_1 + h_2 > 0$, all real z_e, z_h
T3	μ is measurable, continuous and bounded on $(0, \infty)$; the Laplace transform is smooth with $(-1)^r g^{(r)} \geq 0$ for all r (a verbatim copy of the library's `AdmissiblePotential`); the series $\sum_n$ of image terms equals $\int e^{-ts}\mu(s)ds$ for every $t > 0$

No theorem takes the Poisson identity, a summability fact or a measure-theoretic statement as a hypothesis; the only side conditions are $0 < s, t$ and $0 < h_1 + h_2$. The verifier compiled the file (exit 0, no warnings), found no `sorry`/`axiom`/ `native_decide`, checked that the statement matches $V_{ee} + V_{hh} - 2V_{eh}$ with the stated image positions, and recomputed the kernel with its own three-dimensional image sum ($N = 4000$): agreement with the Lean-side expression to 10^{-9} at three radii and with the Laplace integral to 10^{-12}.

What it does not do. It does not prove that 0 is excluded (the zero function is admissible), does not prove uniqueness of the Green function, and does *not* prove the lattice-optimality theorem of [2]: the Lean file shows the *hypothesis* of that theorem is met by the gated kernel, which is as far as the dependency goes. The layered-dielectric extension (a Sturm–Liouville expansion) remains a remark, not part of this file.

3 A harmonic estimate of the crystallisation threshold

Question. At what exciton density does a dual-gated bilayer crystallise? A quantum-Monte-Carlo (QMC) answer needs a code and compute that were not used; this package produces a *harmonic Lindemann estimate* with a calibration error attached, not a critical density. Literature gate (arXiv texts read by the worker and re-read by the verifier): the bare $1/r^3$ gas–solid transition $nr_0^2 = 290(30)$ with Lindemann ratio $\gamma = 0.230(6)$ (DMC) [4], $nr_0^2 \approx 324$ (196–484, PIMC at $T = 0.014D/a^3$) [5]; composite excitons with a crystal confined to a finite density window [6]; gated Coulomb electrons where the harmonic and DMC critical gate distances are 35 and 29 (20%) [7]. Dimensionless set-up: $\Lambda = r_0/d = d/a_M$, $\nu = nd^2$, $nr_0^2 = \nu\Lambda^2$, $D = 2h$.

Pre-registered checks and outcomes.

item	result
P0 lattice code vs. $E_\triangle/N = 4.446(nr_0^2)^{3/2}$	4.4464, deviation 8×10^{-5}: pass
P0 Laplace kernel vs. image series	relative difference $\le 2.4 \times 10^{-8}$: pass
P1 harmonic ratio below the DMC one	**refuted**: $\gamma_h(290) = 0.263$ vs. DMC 0.230(6)
P2 criterion A usable (ratio in $[0.5, 2]$)	**failed**: $n_c^h = 847$ (690–1047), ratio 2.9
P4 gates raise the threshold	confirmed
P4 monotone in ν	**refuted** for all bilayer kernels: interior minimum (the $2/x$ core); monotone only for pure $1/r^3$

Calibrated estimate. The harmonic Lindemann ratio follows $\gamma_h = c\,(nr_0^2)^{-1/8}$ with $c = 0.5342$; the verifier reproduced $c = 0.534241$ with its own real-space image series and Brillouin-zone sum. Because $n_c \propto \gamma_c^{-8}$ the criterion is extremely steep ($\gamma_c = 0.25 \to 435$, $0.20 \to 2600$, $0.15 \to 2.6 \times 10^4$), so the generic "$\gamma_c = 0.1$–0.2" is useless here. Calibrating on the QMC control (criterion B, $\gamma_c = \gamma_h(290) = 0.263$) gives, with a 20% transfer error in the controlling quantity assumed from [7]:

- **A rigid-exciton crystal needs a large coupling.** The smallest Λ with any harmonic-stable density is 48.7 ungated (46–51 over the 260–320 anchors), 52 at $D/d = 3$, 65 at 2, 104 at 1.5, 246 at 1.2; for $D/d \ge 5$ the change is below 2% (cover figure, panel b).
- **Gates raise the threshold** at every one of the eight sampled D/d values within this harmonic model (not a theorem). At $\Lambda = 300$ the crystallisation density $n_c r_0^2$ is 294 ungated (reproduced by construction of the calibration), 419 ($D/d = 20$), 944 (10), 2.7×10^3 (5), 6.7×10^3 (3), 1.6×10^4 (2): the finite range of the interaction is the cause.

- **Mapping.** With placeholder values $d = 1$ nm, $M = m_e$, $\epsilon = 4.5$, $\Lambda = 4.2$, a factor 12 below $\Lambda_{\min}$; over $d \in 0.7$–2 nm, $M \in 0.5$–$1.5\,m_e$, $\epsilon \in 3$–6, Λ runs 1.1–19. In this model no density gives a harmonic-stable crystal for such a bilayer; $\Lambda_{\min} = 49$ corresponds to $d \approx 12$ nm at the placeholder mass and permittivity.

Limits stated by the worker and the verifier. (i) The refutation of P1 (0.230 vs. 0.263) depends on the DMC paper's $\langle (r - r_c)^2 \rangle$ being the full two-dimensional displacement; a factor $\sqrt{2}$ in that definition would flip it, and the estimator was not checked. (ii) The 20% transfer error is taken from a different system, where it applies to a gate distance, not to $n_c \propto \gamma_c^{-8}$; the propagated uncertainty in n_c is therefore larger than a naive reading. (iii) No QMC data exist for the gated kernel; nothing here tests the estimate against any. (iv) It gives no first-order transition, no liquid-side information, no exchange, no composite-exciton physics at spacing $\sim d$, no screening by the gates' own response, and no thermal melting temperature. (v) A post-hoc comparison with the composite-exciton PIMC (threshold $d \leq 9.1\,a_B^*$, i.e. $\Lambda \approx 31$ in these units by the worker's conversion) is a consistency remark only.

4 Edge energies of three CM kernels in open containers

What this is. Classical N-particle minimum-energy configurations in hard-wall containers (quadratic penalty $\kappa = 10^5$) for three pair kernels in units $d = 1$, density $\rho_0 = 0.5/d^2$: the ungated dipole kernel "dip" ($2/r - 2/\sqrt{r^2+1}$), and the gated kernel for $D/d = 3$ (g3) and $D/d = 10$ (g10), tabulated from the image series by Chebyshev interpolation and checked against an independent K_0-series. Containers: disk, square, rectangles with aspect ratios 3:2, 2:1, 3:1, equilateral triangle, hexagon, and a harmonic trap (reference: the local-density approximation). The upstream lattice theorem concerns periodic systems; *open geometries are outside its hypotheses* and nothing here tests it.

Instrument gates. E-a (closed-form $N = 1, 2, 3, 7$ trap energies) and E-a2 pass; E-b (instrument against an independent numpy implementation, energies to 10^{-12}, forces, analytic Hessian against finite differences) passes, and planted bugs 1–4 all make E-b fail. The verifier re-evaluated all 65 best configurations with its own image-series/polygon/disk code: interaction energies agree to 10^{-15} (dip) and to the truncation of its own series elsewhere, wall energies to 7×10^{-11}. E-c, a weak conditional lower bound derived from the upstream theorem, detects a halved energy in **0 of 55** hard-wall cases (registered as "reported as measured"): E-c has no teeth and is not a validation. E-e: 58 of 65 best configurations are stationary ($\max |F| < 10^{-9}$, $\lambda_{\min} \geq -10^{-6}$); the other seven are listed as not certified.

Production. Of the 76 registered cases, 65 were run ($N = 19, 37, 61, 91$ for dip; $N = 19, 37, 61$ for g3 and g10 on their registered shapes); g3/g10 at $N = 91$ were not run. The number of starts per case was reduced (registered random+patch starts: 60+20 at $N = 19$, 40+12 at 37, 16+6 at 61, 8+4 at 91); amendments A6/A6b then added 44 extra random starts at $N = 61$ and 32 at $N = 91$ for the *ungated* kernel only, so the ungated kernel has 60+6 and 40+4 starts there while g3 and g10 keep 16+6 and 8+4. This search asymmetry bears on the gated/ungated comparison below, and so the energies at larger N are *upper bounds on the ground state of the penalised problem*, not certified ground states; the success rate (fraction of screened starts reaching the best energy) is 0.00–0.05 for several $N = 61, 91$ cases.

Findings.

- **The edge excess is negative** for hard walls (a boundary particle misses neighbours), as expected for a repulsive kernel.
- **Aspect ratio.** E_{edge} decreases (becomes more negative) and E_{edge}/P, with P the perimeter, increases monotonically with aspect ratio at every N for dip and at $N = 37, 61$ for g3; g3 at $N = 19$ is non-monotone in both (registered output, reported as such). The gated ($D/d = 3$)

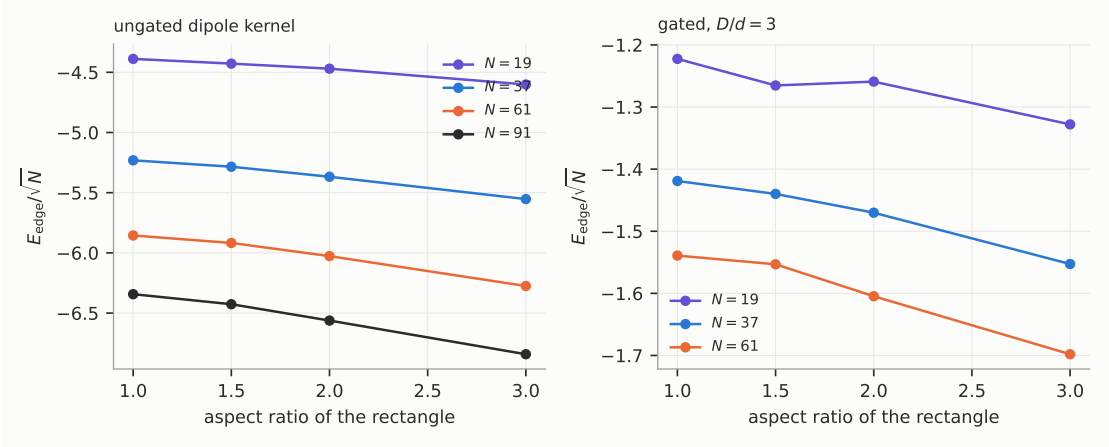


Figure 1: Edge energy per $\sqrt{N}$ against the aspect ratio of the container (best configurations; $E_{\mathrm{edge}} = E_p - Ne_{\mathrm{lat}}(\rho_0)$ with E_p the *penalised* energy). Monotone in the aspect ratio for the ungated kernel at all four N and for g3 at $N = 37, 61$; g3 at $N = 19$ is non-monotone (the second point lies below the third).

edge energy per $\sqrt{N}$ is smaller in magnitude by a factor of 3.4–4.0 than the ungated one at the same N and shape.

- **No clean boundary law.** Fits $E_{\mathrm{edge}} = \gamma\sqrt{N} + c$ have c large and positive (+15 to +19 for dip) and $E_{\mathrm{edge}}/\sqrt{N}$ is not constant (cover figure, panel c); with four ($N = 19 \ldots 91$) or three points the spread of $\gamma_P = \gamma/(P/\sqrt{N})$ across shapes is 12% (dip) and 4.5% (g3); g10 exists on three shapes only. This is a description, not a test of a perimeter law, and the verifier asked that the earlier phrase "holds roughly" not be used.
- **Defects.** Counts of interior five- and seven-fold sites are in the committed table (65 cases). Delaunay counts are unreliable in near-square arrangements (four co-circular points); cases flagged irregular have an irregularity fraction 0.3–1.0, and are not interpreted. A claim, made by the worker's first message, that a relaxed perfect hexagon patch lies 0.16%, 1.2%, 0.36% above the best configuration at $N = 37, 61, 91$ was **not reproducible by the verifier and is withdrawn**; in the committed records the lowest hexagon states at $N = 37, 61, 91$ for the ungated kernel all contain interior defects.

Process weaknesses (registered, not cured). Amendments A4 and A5 (aligned starts) carry timestamps that precede their commit; the aligned stage for $N \leq 37$ had already run when they were committed, and A5 was motivated by looking at success rates, and A6/A6b (extra random starts for the ungated kernel at $N = 61, 91$) were added after seeing low success rates. The Rust source was committed together with the results, not before them (the pre-registration and instrument gates were). Interior-defect definition changed once (A1, before any count was read).

5 A validity gate for Hugging Face and neural-network inputs

Question. Can a Hugging Face dataset or a graph-neural-network (GNN) force field supply the inputs of the gated kernel: the electron–hole separation d (hence $\tilde{U}(0) = 4\pi d(1 - d/D)$), the dielectric environment, the exciton mass and binding energy? **Verdict: the gate fails for the two quantities that matter.**

gate	result	evidence
G1 source: read-only, <50 MB, permissive, hashed	pass	`colabfit/JARVIS_C2DB`, CC-BY-4.0, 1.99 MB
G2 2H $MoSe_2$ and WSe_2 monolayers present	pass	entries 3150 and 3118 identified by geometry
G3 any $MoSe_2/WSe_2$ bilayer so d can be read	**fail**	no entry has two metal planes > 5 Å apart; ≤ 12 atoms
G4 in-plane a within 2% of ≥ 2 published values	pass	3.3197, 3.3190 Å; largest deviation 1.13%; agrees with PBE values [8] by construction
G5 any excitonic/optical quantity	**fail**	only PBE energy and PBE gap

The verifier re-downloaded the files (hashes identical), re-ran the extraction (byte-identical outputs) and re-read the literature values. Corrections it required, now applied: the claim "no Hugging Face dataset or GNN supplies d" is limited to *the datasets examined* (one opened; other JARVIS 2D sets were scanned by the verifier and contain no pure $MoSe_2/WSe_2$ bilayer, but the report had not shown that); the G5 gate is hard-coded to fail in the script, with the null columns in a reason string; and the machine-learned potentials were not run. A ground-state model can in principle supply d (it is a ground-state quantity), but the literature already supplies it to about 1.5%: metal–metal distances (the worker's reading of the figures) of 6.42–6.51 Å for commensurate stackings and 7.10 Å for AA (PBE+D3 [9], LDA [10]); the binding energies in the literature read (0.161–0.55 eV [11, 12, 10]; factor 3.4, mixing intralayer excitons in different environments and methods with interlayer values of 0.25–0.32 eV, factor about 1.3) depend on method and environment, which is the uncertainty an input would have to resolve and which a ground-state model cannot. Conclusion: no neural-network surrogate is warranted at this stage; a PBE value of d is worth less than the spread in ϵ and E_b.

6 ETDRK4 for the driven-dissipative model

What was done. A pure-Rust radix-2 FFT with a fixed-step ETDRK4 (Cox–Matthews [13], contour-integral φ-functions) for the Phase 3 model, with the reservoir decay and the kinetic term in the linear part and the rate term $R|\psi|^2 n_R$ in the nonlinearity; CVODE comparisons use the pinned rusty-SUNDIALS tree. Gates T-a..T-g, amendments A1–A3 dated and committed before the runs they announce.

gate	outcome	reason
T-a steady states	pass	13/13 above threshold, worst $\lvert n-10\rvert/10 = 2.8\times10^{-13}$; 10/10 below ($\lvert\psi\rvert \leq 2\times10^{-18}$)
T-b order 4	**fail**	successive ratios 2.2, 2.9, 4.5, 7.2 (registered 12–20); consistent with stiff-order reduction
T-c stability limit	measured	$h_{\text{first}} = 0.64$ at the registered point, 37× the IMEX 0.01707; 0.0034 at $Rn_0/\gamma_R = 5$
T-d(i) records, $t = 4000$	**fail** for $V_0 = 0.5, 1$	ETD time-step error; converges to the CVODE record as $h \to 0$
T-d(ii) $t = 16000$	2 of 4 pass	$V_0 = 0.5, 1$ fail by 2–3.6× on the relative mean n at $h = 0.05$
T-d(iii) 16×16 vs. the independent reviewer's printed digits (review of paper I)	pass 3/3	shifts $0.0463, 0.0767, 0.1065$; $n_c = 5.844, 2.377, 0.1000$
T-e(ii) time to residual 10^{-7}, 8×8	7.7 s vs. 227 s	factor 29 over dense CVODE, same stationarity time
T-f planted bugs	caught	B2 caught only by the non-stiff control

Why T-b fails. A control with γ_R changed at constant $n = 10$ ($P = \gamma(10 + \gamma_R)$) shows the step-halving error ratio rising towards 16 as $h\gamma_R$ falls: $7.6, 10.9, 13.3, 14.6$ at $\gamma_R = 20$; 19.9

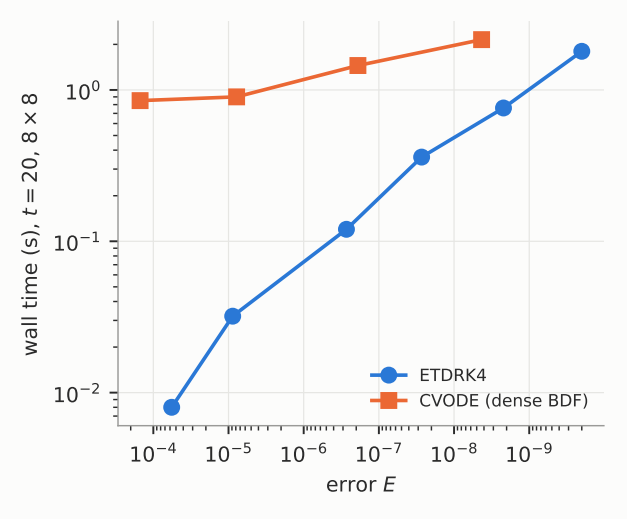


Figure 2: Work–precision, 8×8 grid, $t = 20$. ETD is 20–30× cheaper than dense BDF at errors $\sim 10^{-5}$, about 2.5× at 3×10^{-9}; no crossing was measured. CVODE wall times are the midpoints of the quoted ranges (the machine load was 4–11 on 8 cores).

at $\gamma_R = 2$ (the later ratios sit at the reference floor); 2.2–7.2 at the registered $\gamma_R = 200$, i.e. an observed order of about 1.1–1.5 at the largest steps. Krogstad's variant (cover panel d) has 5–15 times smaller error at $h = 0.0125$ but no order 4 either. The verifier reproduced the ratios with its own ETDRK4 and a DOP853 reference $(2.25, 2.90, 4.49, 7.21)$, which excludes a coding error. The stiff-order reduction of the Cox–Matthews scheme [14] is the mechanism consistent with these data; the citation is from memory and the claim "is exactly the stiff-order reduction" is stronger than a single scan proves.

Where ETD is worse than the alternatives. (i) Its advantage over dense BDF shrinks from 20–30× at $E \sim 10^{-5}$ to about 2.5× at 3×10^{-9} and the tabulated runs do not show a crossing (at $E = 2 \times 10^{-10}$ ETD took 1.8 s, the same wall time as the most accurate CVODE setting, which reached only 4×10^{-9}); the step count relative to CVODE grows with accuracy, so a crossing below 10^{-9} is a suspicion, not a measurement. (ii) Its step limit is not unconditional: kinetic resonances at $hw_k = 2\pi n$ and, at $Rn_0/\gamma_R = 5$, a limit 0.0034, five times below the IMEX module's. (iii) It carries a time-step residual floor $\sim h^3$ in its stationary state (7×10^{-9} at $h = 0.05$), so a gate with budget 10^{-8} needs a smaller step. (iv) It has no error control. The verifier's non-blocking notes: the T-d(iii) tolerances are about the rounding of the reviewer's three-digit values (a weak consistency check); the source was committed with the results; one summary row conflates last-stable and first-failing steps.

7 What was not completed

Not reported here. (a) The matrix-free Newton–Krylov solver (SDIRK2 with GMRES) pre-registered gates K-a..K-e were being run on small and 64×64 grids when this paper was frozen; no Krylov result is claimed. (b) The classical L6 node-arrangement study with the `rusty-Fenris-CMRR` engine and the gated kernel had gates a, b, f and a production campaign, and is awaiting its verifier; no result is claimed, and the open-boundary study of Section 4 is not that study. They may be reported in a later record; nothing is implied here about their outcome.

8 Response to the external review's four improvements

proposal	status in this deposit
Lean formalisation	done to the stated extent (Section 2); the review's claim that Mathlib lacks Poisson summation was checked against the pinned tree and is false for this release
JFNK / Krylov at $\geq 64 \times 64$	not complete; see Section 7
Exponential integrators	done: fails the registered order gate at the registered stiffness; useful as a fast stationary-state solver (Section 6)
L6 with Fenris-CMRR	not complete; a classical design target at most, never a prediction about excitons

9 Limitations and reproducibility

Limitations. The lattice theorem is unrefereed and AI-generated; the gated kernels are ideal grounded planes; the model parameters of the driven-dissipative model are dimensionless and chosen for stiffness; the Wigner estimate is a harmonic criterion calibrated on one QMC anchor; the edge study is classical, with reduced numbers of starts; the HF gate examined one dataset in depth; the ETD tests use 8×8 and 16×16 grids and one initial seed for most gates. The pre-registered protocol L1–L6 of the first paper is unchanged; L6 remains a classical design target.

Files. Lean: `exploration/exciton/lean/GatedKernel.lean`, `VERIFICATION_GatedKernel.md`. Wigner: `docs/designs/EXCITON_FLUID_WIGNER_GATE.md`, `docs/designs/EXCITON_FLUID_WIGNER_PREREG.md`, `docs/designs/EXCITON_FLUID_WIGNER_REPORT.md`, `exploration/exciton/results/wigner/`. Edge: `exploration/exciton/edge/` (`PREREG.md`, `results/analysis/cases.md`, `summary.json`). HF gate: `exploration/exciton/hf_gate/`. ETD: `docs/designs/EXCITON_FLUID_ETD_PREREG.md`, `docs/designs/EXCITON_FLUID_ETD_RESULTS.md`, `exploration/exciton/rust/qf-exciton-etd/`. Response to the external review: `docs/designs/EXCITON_FLUID_GATED_PEER_REVIEW_RESPONSE.md`. Figures: `exploration/exciton/python/companion_figures.py`.

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